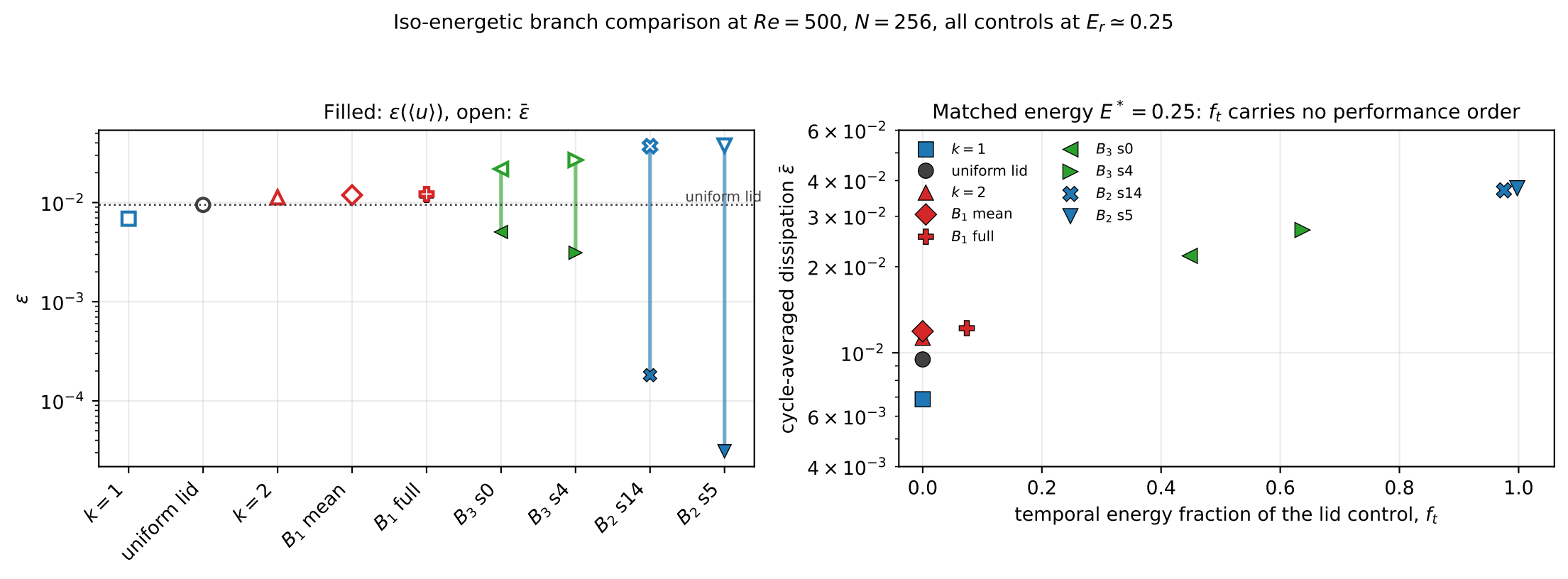
Iso-energetic branch comparison at $Re = 500$, $N = 256$, all controls at $E_r \simeq 0.25$
Filled: $\varepsilon(\langle u \rangle)$, open: $\bar{\varepsilon}$
ε
10^{-2}
10^{-3}
10^{-4}
uniform lid
$k = 1$
uniform lid
$k = 2$
B_1 mean
B_1 full
B_3 s0
B_3 s4
B_2 s14
B_2 s5
Matched energy $E^* = 0.25$: f_t carries no performance order
cycle-averaged dissipation $\bar{\varepsilon}$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
$k = 1$
uniform lid
$k = 2$
B_1 mean
B_1 full
B_3 s0
B_3 s4
B_2 s14
B_2 s5
0.0
0.2
0.4
0.6
0.8
1.0
temporal energy fraction of the lid control, f_t